

Homework #2

Conditional MI, Relative Entropy, and Inequalities

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw2.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw2.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW2.1 (C&T 2.6) *Conditional mutual information vs. unconditional mutual information.* Give examples of joint random variables X , Y , and Z such that

- (a) $I(X; Y | Z) < I(X; Y)$.
- (b) $I(X; Y | Z) > I(X; Y)$.

Problem HW2.2 (C&T 2.23) *Conditional mutual information.* Consider a sequence of n binary random variables $X_1, X_2, \dots, X_n$. Each sequence with an even number of 1's has probability $2^{-(n-1)}$, and each sequence with an odd number of 1's has probability 0. Find the mutual informations

$$I(X_1; X_2), \quad I(X_2; X_3 | X_1), \quad \dots, \quad I(X_{n-1}; X_n | X_1, \dots, X_{n-2}).$$

Problem HW2.3 (C&T 2.26) *Another proof of nonnegativity of relative entropy.* In view of the fundamental nature of the result $D(p||q) \geq 0$, we will give another proof.

- (a) Show that $\ln x \leq x - 1$ for $0 < x < \infty$.
- (b) Justify the following steps:

$$-D(p||q) = \sum_x p(x) \ln \frac{q(x)}{p(x)} \tag{2.176}$$

$$\leq \sum_x p(x) \left(\frac{q(x)}{p(x)} - 1 \right) \tag{2.177}$$

$$\leq 0. \tag{2.178}$$

- (c) What are the conditions for equality?

Problem HW2.4 (C&T 2.29) *Inequalities.* Let X , Y , and Z be joint random variables. Prove the following inequalities and find conditions for equality.

- (a) $H(X, Y | Z) \geq H(X | Z)$.
- (b) $I(X, Y; Z) \geq I(X; Z)$.

- (c) $H(X, Y, Z) - H(X, Y) \leq H(X, Z) - H(X)$.
- (d) $I(X; Z | Y) \geq I(Z; Y | X) - I(Z; Y) + I(X; Z)$.

Problem HW2.5 (C&T 2.34) *Entropy of initial conditions.* Prove that $H(X_0 | X_n)$ is nondecreasing with n for any Markov chain.

Problem HW2.6 (C&T 2.40) *Discrete entropies.* Let X and Y be two independent integer-valued random variables. Let X be uniformly distributed over $\{1, 2, \dots, 8\}$, and let $P\{Y = k\} = 2^{-k}$, $k = 1, 2, 3, \dots$

- (a) Find $H(X)$.
- (b) Find $H(Y)$.
- (c) Find $H(X + Y, X - Y)$.

Problem HW2.7 (C&T 2.43) *Mutual information of heads and tails.*

- (a) Consider a fair coin flip. What is the mutual information between the top and bottom sides of the coin?
- (b) A six-sided fair die is rolled. What is the mutual information between the top side and the front face (the side most facing you)?

Problem HW2.8 (C&T 2.48) *Sequence length.* How much information does the length of a sequence give about the content of a sequence? Suppose that we consider a Bernoulli ($\frac{1}{2}$) process $\{X_i\}$. Stop the process when the first 1 appears. Let N designate this stopping time. Thus, X^N is an element of the set of all finite-length binary sequences $\{0, 1\}^* = \{0, 1, 00, 01, 10, 11, 000, \dots\}$.

- (a) Find $I(N; X^N)$.
- (b) Find $H(X^N | N)$.
- (c) Find $H(X^N)$.

Let's now consider a different stopping time. For this part, again assume that $X_i \sim \text{Bernoulli}(\frac{1}{2})$ but stop at time $N = 6$, with probability $\frac{1}{3}$ and stop at time $N = 12$ with probability $\frac{2}{3}$. Let this stopping time be independent of the sequence $X_1 X_2 \dots X_{12}$.

- (d) Find $I(N; X^N)$.
- (e) Find $H(X^N | N)$.
- (f) Find $H(X^N)$.